🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.1

Area of Regular Polygons

Lesson 5-9 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the area of a regular polygon by decomposing it into triangles.
Language Objective: I can explain my steps using the words regular polygon, decompose, triangle, and composite.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Area of Regular Polygons | **Regular Polygon** | **Decompose** | **Triangle** | **Composite** | **Formula**

👆 *Tap any word to see what it means and a picture.*

1. The space inside a regular polygon, found with A = (½bh) × n, is the ______________.

2. A polygon with all sides and all angles equal is a ______________.

3. To break a shape into smaller shapes is to ______________ it.

4. A three-sided polygon is a ______________.

5. Made of two or more shapes combined, a figure is ______________.

6. A math rule written with symbols is a ______________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How much will the gazebo flooring cost?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Area of Regular Polygons** — The total space a regular polygon covers, found by splitting it into congruent triangles from the center. Formula: A = (½bh) × n — one triangle's area times the number of triangles.
  Área de polígonos regulares — El espacio total que cubre un polígono regular, hallado al dividirlo en triángulos congruentes desde el centro. Fórmula: A = (½bh) × n — el área de un triángulo por el número de triángulos.

- [ ] **Regular Polygon** — A closed flat shape made of straight sides, where every side is the same length and every angle is the same size.
  Polígono regular — Una figura plana cerrada de lados rectos, donde todos los lados miden lo mismo y todos los ángulos son iguales.

- [ ] **Decompose** — To break a shape into smaller, simpler shapes.
  Descomponer — Separar una figura en figuras más pequeñas y simples.

- [ ] **Triangle** — A shape with three sides.
  Triángulo — Una figura de tres lados.

- [ ] **Composite** — Made by putting two or more simple shapes together.
  Compuesto — Formada al juntar dos o más figuras simples.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

One triangle's area is ____ sq ft. The octagon has ____ triangles, so total area = ____ sq ft. The cost is ____ x $3 = $____.
Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: We decompose the hexagon into triangles because we already know how to find a triangle's area.

Evidence: Students explain that splitting the hexagon into 6 equal triangles lets them use the triangle area formula they already know, then combine the parts.

Reasoning: This evidence connects the definition of area of regular polygons to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **One triangle's area is ____ sq ft. The octagon has ____ triangles, so total area = ____ sq ft. The cost is ____ x $3 = $____.**
- **My answer is ____.**

  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **I know because ____.**

  Lo sé porque ____.
- **So ____.**

  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
  Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Area of Regular Polygons
2. **Notes 2:** Regular Polygon
3. **Notes 3:** Decompose
4. **Notes 4:** Triangle
5. **Notes 5:** Composite
6. **Notes 6:** Formula
7. **Watch for this mistake:** *A common mistake in Area of Regular Polygons is stopping after finding the area of ONE triangle and reporting that as the total area — or adding the number of triangles instead of multiplying. Remember: total area = (one triangle's area) × (number of sides), not (one triangle's area) + (number of sides).*
8. **Try It 1:** A. 60 in^2 — *One triangle = ½ × 4 × 6 = 12 in^2. A pentagon has 5 triangles, so total = 5 × 12 = 60 in^2.*
9. **Try It 2:** A. 120 sq ft — *An octagon has 8 sides, so it splits into 8 triangles. One triangle = ½ × 6 × 5 = 15 sq ft. Total = 8 × 15 = 120 sq ft.*